

Passaggio parametri versione n. 1

- Scoping dinamico, MODE = OUT per riferimento
- Scoping statico, MODE = IN OUT per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([IN OUT x rif] int a)
    int d; int b;
    procedure p3([IN OUT x rif] int b)
      int d;
      procedure p4([IN OUT x rif] int d,[MODE] int c)
        int b;
        BEGIN
          b=c-3;
          if c>7 then d=d*3 else d=4;
          c=4;
          a=c;
          write(a,b,c,d);
        END

        BEGIN
          d=2;
          if c>5 then b=2 else b=1;
          c=b+4;
          a=3;
          p4(a, c);
          write(a,b,c,d);
        END

        BEGIN
          d=c-3;
          b=1;
          a=d;
          c=b;
          p3(c);
          write(a,b,c,d);
        END

        BEGIN
          a=3;
          b=2;
          c=2;
          d=3;
          p2(b);
          write(a,b,c,d);
        END
```

Passaggio parametri versione n. 2

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN x copia] int s)
    int r; int q;
    BEGIN
      r=2;
      q=s;
      s=q+3;
      t=s-1;
      write(q,r,s,t);
    END

    procedure p3([MODE] int t)
      int r;
      BEGIN
        r=s*3;
        t=s;
        q=r*2;
        s=s;
        p4(s);
        write(q,r,s,t);
      END

      procedure p4([IN OUT x rif] int s)
        int r; int q;
        BEGIN
          r=t-4;
          q=1;
          s=q*3;
          t=1;
          p2(q);
          write(q,r,s,t);
        END

      BEGIN
        q=2;
        r=2;
        s=3;
        t=1;
        p3(s);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 3

- Scoping dinamico, MODE = IN OUT per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int x; int y; int z; int t;
procedure p2([IN OUT x copia] int x)
  int t;
  procedure p3([IN x copia] int y,[IN x copia] int t)
    int z;
    procedure p4([IN x copia] int z,[MODE] int y)
      int t;
      BEGIN
        t=y*1;
        z=4;
        if y<4 then y=y+1 else y=x*4;
        if x=9 then x=4 else x=3;
        write(x,y,z,t);
      END

    BEGIN
      z=x;
      y=3;
      t=y-4;
      x=y;
      p4(t, x);
      write(x,y,z,t);
    END

  BEGIN
    t=1;
    x=2;
    z=4;
    y=3;
    p3(x, z);
    write(x,y,z,t);
  END

BEGIN
  x=3;
  y=3;
  z=3;
  t=2;
  p2(z);
  write(x,y,z,t);
END
```

Passaggio parametri versione n. 4

- Scoping dinamico, MODE = OUT per riferimento
- Scoping statico, MODE = IN OUT per riferimento

```
program p1
int q; int r; int s; int t;
  procedure p2([MODE] int t)
    int r; int q;
    BEGIN
      if s=7 then r=1 else r=t-1;
      q=s;
      if s=0 then t=4 else t=2;
      s=1;
      write(q,r,s,t);
    END

    procedure p3([IN OUT x copia] int q,[IN OUT x copia] int t)
      int s;
      BEGIN
        s=3;
        q=t;
        t=q*2;
        r=4;
        p4(r, q);
        write(q,r,s,t);
      END

      procedure p4([IN x copia] int r,[IN OUT x copia] int s)
        int t;
        BEGIN
          t=q+1;
          if t<8 then r=4 else r=r;
          s=2;
          q=3;
          p2(t);
          write(q,r,s,t);
        END

      BEGIN
        q=4;
        r=3;
        s=4;
        t=0;
        p3(r, t);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 5

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN OUT per riferimento

```
program p1
int x; int y; int z; int t;
  procedure p2([IN x copia] int t)
    int z; int x;
    procedure p3([IN OUT x copia] int t)
      int z;
      BEGIN
        if y>2 then z=t*2 else z=4;
        t=y;
        x=4;
        y=1;
        p4(z);
        write(x,y,z,t);
      END

      BEGIN
        z=1;
        x=2;
        t=x+1;
        y=2;
        p3(x);
        write(x,y,z,t);
      END

    procedure p4([MODE] int z)
      int x; int t;
      BEGIN
        x=3;
        t=3;
        z=3;
        y=z;
        write(x,y,z,t);
      END

    BEGIN
      x=2;
      y=1;
      z=2;
      t=2;
      p2(t);
      write(x,y,z,t);
    END
```

Passaggio parametri versione n. 6

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN OUT x rif] int r,[IN OUT x rif] int t)
    int s;
      procedure p3([IN OUT x rif] int t,[IN OUT x copia] int q)
        int r;
        BEGIN
          r=s;
          t=1;
          q=s;
          s=q+3;
          p4(q);
          write(q,r,s,t);
        END

      procedure p4([MODE] int t)
        int r; int s;
        BEGIN
          r=t-3;
          s=q+2;
          t=r+1;
          if t<8 then q=r-3 else q=t+3;
          write(q,r,s,t);
        END

      BEGIN
        s=4;
        r=q;
        t=4;
        if r<5 then q=q else q=4;
        p3(q, t);
        write(q,r,s,t);
      END

  BEGIN
    q=3;
    r=3;
    s=0;
    t=4;
    p2(t, r);
    write(q,r,s,t);
  END
```

Passaggio parametri versione n. 7

- Scoping dinamico, MODE = OUT per riferimento
- Scoping statico, MODE = IN per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([MODE] int z,[IN OUT x rif] int t)
    int y;
    BEGIN
    if t>10 then y=2 else y=t*1;
    t=t-2;
    x=3;
    p4(t);
    write(x,y,z,t);
    END

    procedure p3([IN OUT x rif] int y,[IN OUT x copia] int z)
      int t;
      BEGIN
      t=4;
      if x<8 then y=3 else y=t;
      z=y;
      x=4;
      write(x,y,z,t);
      END

      procedure p4([IN OUT x rif] int y)
        int t; int z;
        BEGIN
        t=1;
        z=x+2;
        if x<6 then y=x else y=2;
        x=3;
        p3(y, z);
        write(x,y,z,t);
        END

    BEGIN
    x=1;
    y=0;
    z=0;
    t=1;
    p2(x, z);
    write(x,y,z,t);
    END
```

Passaggio parametri versione n. 8

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per riferimento

```
program p1
int q; int r; int s; int t;
procedure p2([IN OUT x copia] int q)
  int r;
    procedure p3([MODE] int t,[IN x copia] int r)
      int s;
      BEGIN
        s=3;
        t=2;
        r=2;
        q=1;
        p4(q, r);
        write(q,r,s,t);
      END

    procedure p4([IN OUT x rif] int q,[IN OUT x copia] int t)
      int s;
      BEGIN
        s=3;
        q=4;
        t=2;
        r=q*1;
        write(q,r,s,t);
      END

  BEGIN
    r=t;
    q=2;
    if t<2 then s=2 else s=r*3;
    if r>8 then t=3 else t=2;
    p3(t, q);
    write(q,r,s,t);
  END

BEGIN
q=4;
r=0;
s=1;
t=4;
p2(t);
write(q,r,s,t);
END
```


Passaggio parametri versione n. 9

- Scoping dinamico, MODE = OUT per riferimento
- Scoping statico, MODE = IN OUT per riferimento

```
program p1
int q; int r; int s; int t;
  procedure p2([IN x copia] int t)
    int r; int s;
    BEGIN
      r=2;
      s=q*4;
      t=4;
      q=r;
      p3(t, s);
      write(q,r,s,t);
    END

    procedure p3([MODE] int r,[IN OUT x copia] int q)
      int t;
      BEGIN
        if r=3 then t=r else t=r-2;
        r=1;
        q=s;
        s=2;
        write(q,r,s,t);
      END

      procedure p4([IN x copia] int s)
        int q; int t;
        BEGIN
          q=r;
          t=1;
          if r>6 then s=t else s=4;
          r=2;
          p2(t);
          write(q,r,s,t);
        END

      BEGIN
        q=3;
        r=4;
        s=4;
        t=1;
        p4(q);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 10

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = IN per copia

```
program p1
int x; int y; int z; int t;
  procedure p2([MODE] int x)
    int t; int y;
      procedure p3([IN OUT x rif] int y)
        int t; int z;
        BEGIN
          t=3;
          z=y-3;
          y=3;
          x=z;
          write(x,y,z,t);
        END

      procedure p4([IN OUT x copia] int z)
        int x; int t;
        BEGIN
          x=y;
          t=y-3;
          if t=3 then z=z else z=t;
          if t<1 then y=z else y=x+3;
          p3(z);
          write(x,y,z,t);
        END

    BEGIN
      t=1;
      if z<5 then y=4 else y=4;
      x=4;
      z=x;
      p4(t);
      write(x,y,z,t);
    END

  BEGIN
    x=2;
    y=4;
    z=1;
    t=1;
    p2(t);
    write(x,y,z,t);
  END
```

Passaggio parametri versione n. 11

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN OUT x rif] int t)
    int q; int r;
    BEGIN
      q=s;
      r=2;
      t=s;
      s=t;
      p3(r);
      write(q,r,s,t);
    END

    procedure p3([IN x copia] int r)
      int t;
      BEGIN
        t=q*4;
        r=4;
        q=2;
        if r<0 then s=t else s=q;
        write(q,r,s,t);
      END

      procedure p4([IN OUT x copia] int t,[MODE] int r)
        int s;
        BEGIN
          s=4;
          t=s;
          r=2;
          q=s*1;
          p2(r);
          write(q,r,s,t);
        END

      BEGIN
        q=3;
        r=3;
        s=0;
        t=3;
        p4(t, s);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 12

- Scoping dinamico, MODE = OUT per copia
- Scoping statico, MODE = IN OUT per copia

```
program p1
int q; int r; int s; int t;
procedure p2([IN OUT x rif] int t)
  int q; int s;
  procedure p3([MODE] int r)
    int s; int t;
    procedure p4([IN x copia] int s)
      int t;
      BEGIN
        t=q;
        if s=20 then s=1 else s=r+3;
        q=1;
        r=2;
        write(q,r,s,t);
      END

      BEGIN
        if r=10 then s=2 else s=1;
        t=r;
        if q<5 then r=4 else r=s*4;
        if r=1 then q=r else q=2;
        p4(q);
        write(q,r,s,t);
      END

      BEGIN
        q=r;
        s=2;
        t=q;
        r=t;
        p3(t);
        write(q,r,s,t);
      END

      BEGIN
        q=4;
        r=2;
        s=1;
        t=2;
        p2(r);
        write(q,r,s,t);
      END
```

Passaggio parametri versione n. 13

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([IN OUT x copia] int b,[IN OUT x copia] int c)
    int d;
    BEGIN
    d=b;
    b=b+3;
    c=3;
    a=d+3;
    write(a,b,c,d);
    END

  procedure p3([IN OUT x rif] int c)
    int a;
    procedure p4([MODE] int c)
      int b; int d;
      BEGIN
      if a>20 then b=3 else b=2;
      d=3;
      c=b;
      a=2;
      p2(a, d);
      write(a,b,c,d);
      END

    BEGIN
    a=d*3;
    c=3;
    b=c;
    d=4;
    p4(d);
    write(a,b,c,d);
    END

  BEGIN
  a=0;
  b=2;
  c=2;
  d=0;
  p3(a);
  write(a,b,c,d);
  END
```

Passaggio parametri versione n. 14

- Scoping dinamico, MODE = IN per copia
- Scoping statico, MODE = OUT per copia

```
program p1
int a; int b; int c; int d;
  procedure p2([MODE] int b,[IN OUT x copia] int c)
    int a;
    BEGIN
      a=b+1;
      b=4;
      c=2;
      d=3;
      p3(b, d);
      write(a,b,c,d);
    END

    procedure p3([IN OUT x copia] int c,[IN OUT x rif] int a)
      int b;
      procedure p4([IN OUT x rif] int d,[IN x copia] int b)
        int c;
        BEGIN
          c=3;
          d=a+2;
          b=3;
          a=3;
          write(a,b,c,d);
        END

        BEGIN
          b=a;
          c=1;
          a=1;
          if a>9 then d=2 else d=d*4;
          p4(c, d);
          write(a,b,c,d);
        END

      BEGIN
        a=4;
        b=4;
        c=4;
        d=0;
        p2(c, b);
        write(a,b,c,d);
      END
```

Passaggio parametri versione n. 15

- Scoping dinamico, MODE = IN per riferimento
- Scoping statico, MODE = OUT per copia

```
program p1
int q; int r; int s; int t;
  procedure p2([IN OUT x copia] int t)
    int r;
    BEGIN
      r=3;
      t=q;
      q=1;
      s=r+1;
      p4(s, r);
      write(q,r,s,t);
    END

    procedure p3([MODE] int s)
      int q;
      BEGIN
        q=t;
        s=r*3;
        if r<8 then t=r-4 else t=2;
        r=t*3;
        write(q,r,s,t);
      END

      procedure p4([IN x copia] int q,[IN OUT x rif] int r)
        int t;
        BEGIN
          t=r*1;
          q=3;
          r=r*3;
          if r=7 then s=2 else s=s-2;
          p3(r);
          write(q,r,s,t);
        END

      BEGIN
        q=1;
        r=2;
        s=4;
        t=0;
        p2(t);
        write(q,r,s,t);
      END
```